

Charge Controller QFD Chart

Design Specification	Spec.	Impact	Out source	Fabricate	Assemble	Wave Solder	Hand solder	PHP Program	Embedded Prog	MQTT Prog	Web site	Embedded CGI
Power Driver	Relay	23	1									
CPU	A ProMini	30	1		1							
Connectivity	ESP8266 WiFi	18	1		1							
Enclosure	mini Euro IP65	17	1		1							
Layout Plan	Modular	7		1								
Data Storage	SD Card, Cloud(Opt)	19			1				1			
Warranty	1 yr	16	1									
Env Norm	Lead free	12				1	1					
Trade practice	Fair trade	7	1	1	1	1	1					
Current Sensor	0.02 ohm resistor	4	1									
Volt sensor	resistor bridge, CPU	4	1									
Temp/Humid Sensor	DHT 11	8	1									
Weather Sensors	Rain, wind (opt)	4	1									
Cloud storage	Optional	10	1					1		1	1	1
Analytics	Cloud based	5						1				
Importance			141	14	91	19	19	15	19	10	10	10

Fig. 5: Third house—Manufacturing plan

Charge Controller QFD Chart

Mfg. Plan	Impact	Incoming Inspection	Process Monitoring	Poka Yoke	Sample inspection
Out source	141	1			
Fabricate	14		1		
Assemble	91			1	
Wave Solder	19		1		
Hand solder	19		1		
PHP Program	15		1		
Embedded Prog	19		1		
MQTT Prog	10				1
Web site	10				1
Embedded CGI	10				1
Importance		141	86	91	30

Fig. 6: Fourth House—Quality Assurance plan

stage give better focus on the design process.

In the second house, we list the high-level design parameters in the columns and then map these parameters to product specifications. In the end, we examine the co-relations between design parameters and decide on the design parameters.

Design specifications and the decided specifications we list in the third house, the manufacturing plan. Design and manufacturing plan must emerge together. One needs to

look at various alternatives, work out production volume, and then decide on the most profitable manufacturing route. The cost of the part as well as the production costs have to be worked out before we take the design decision.

The charge controller needs to handle rather a large current. The frequency of switching on/off is less. All the competitions use a relay for this operation. We also decide on a relay for the main current driver operation. In our device, we need to sense two analogue voltages and get data from the humidity and temperature sensor using an i2C connection.

As the expected production volume of our product is small to medium, we select a readily available microcontroller module, namely, Arduino Pro-Mini. For internet connectivity, we select a companion Wi-Fi chip, ESP8266. Likewise, we decide on enclosure, sensors, environment norms, and software options. These details are put in the third QFD chart (Fig. 5).

The third chart help us prepare the manufacturing plan for our design. Here we put the manufacturing

process in the columns and figure out the manufacturing processes for various parts. In the impact column, we use the figures from the second QFD chart. Summation of the impact factors gives us the importance of manufacturing processes. Manufacturing processes that have a high impact on the product need close monitoring to ensure quality.

In the fourth house, we decide on the quality assurance plan for our product. Here we decide how to monitor the process and ensure the quality of the output. As you see in Fig. 6, we may adopt a similar quality control strategy for multiple production processes.

Like our previous step, the importance value of stage 3 becomes the impact of stage 4. The summation of relevant impact factors gives importance to the quality control process. A quality process with high importance value, like the incoming inspection in our case, needs to be given more focus. We need to develop elaborate plans to ensure proper control.

These four QFD charts give us conceptual clarity of our product design. QFD process ensures that all stages of product design, manufacturing, and quality assurance stay aligned with the customer's requirements.

What we got after this exercise is the base-level product concept. It is a framework on product requirements, basic design parameters, a conceptual clarity of various manufacturing processes required, and the quality assurance process. It is not the final product design.

The final product design will involve further detailing. It will also require analysis of cost, technology, and business viability. Finally, product and business validation has to happen. In our next articles, we shall explain these processes that are needed for successful product design. **EFY**